Collection: Lab 1:
Group No. 23

Instructor's Signature Sudipto 5/2/26

Table 1:

	Resistance using colour coding					Resistance using DMM	% of Error
	Band 1	Band 2	Band 3	Band 4	Resistance ± tol		
4.7 kΩ	Yellow 4	Violet 7	Red 2 (10^2)	Gold 5%	4.7k ±5%	4.53 kΩ	1.49%
10 kΩ	Brown 1	Black 0	Orange 3 (10^3)	Gold 5%	10k ±5%	9.85 kΩ	1.5%

Table 2: Experimental readings for 4.7 kΩ

Supply V_{in}	Voltage Measurement Voltage, V_R (V)	Current Measurement Current, I (mA)	Power Calculation Power, VI (mW)
2V	1.98	0.43	0.8514
4V	3.95	0.85	3.3575
6V	5.98	1.29	7.7142
8V	7.99	1.72	13.7428
10V	9.99	2.14	21.3786

Table 3: Experimental readings for 10 kΩ

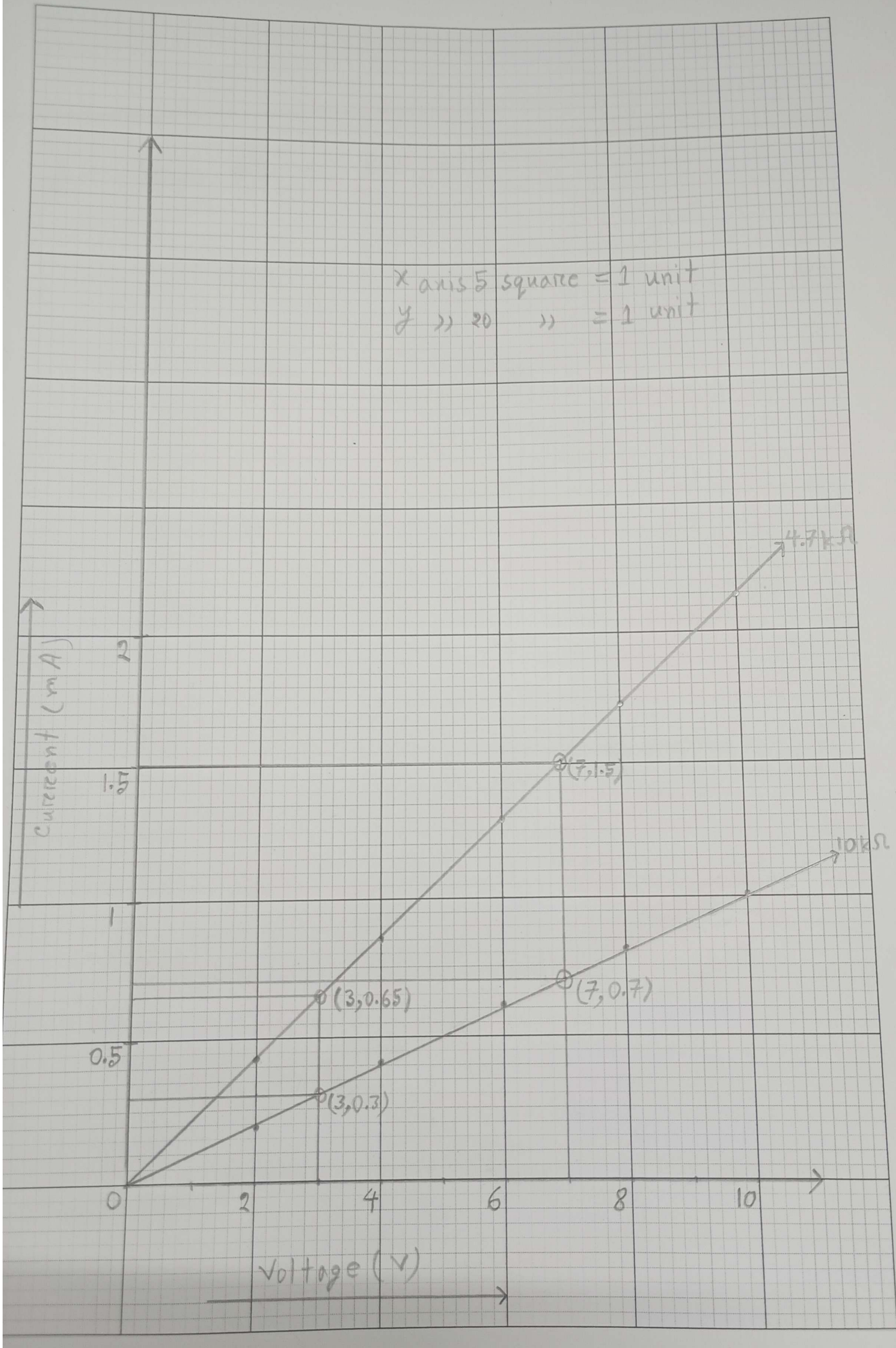
Supply V_{in}	Voltage Measurement Voltage, V_R (V)	Current Measurement Current, I (mA)	Power Calculation Power, VI (mW)
2V	1.95	0.19	0.3705
4V	3.93	0.40	1.572
6V	5.98	0.61	3.6478
8V	8.05	0.81	6.5205
10V	10.03	1.01	10.130

"If the circuit works, means you've just mastered a building block of modern tech"

Questions

1. Plot V (voltage) vs. I (current) graph for each resistor value in same graph.
2. Does your experimental circuit follow ohm's law? Explain how you figured it out.
3. Calculate the resistance of each circuit using the slope of your V vs. I graph. Compare these R graph values to the measured R values using DMM. Find the percent difference.
4. Based on Ohm's law, describe the linear relationship between voltage and current.

MultiSim Simulation


x axis 5 square = 1 unit
y ›› 20 ›› = 1 unit
4.7kΩ
10kΩ
(7,1.5)
(3,0.65)
(7,0.7)
(3,0.3)
2
1.5
1
0.5
0
2
4
6
8
10
Current (mA)
Voltage (V)

Slope for 4.7 kΩ $= \frac{Y_2 - Y_1}{X_2 - X_1} = \frac{1.5 - 0.65}{7 - 3} = 0.2125$

$\therefore$ Resistance $= \frac{1}{\text{Slope for } 4.7\,k\Omega}$

$= \frac{1}{0.2125}$

$= 4.706\ k\Omega$

Measured R was $= 4.63\ k\Omega$

$\therefore$ Error $= \frac{4.706 - 4.63}{4.706} \times 100\%$

$= 1.61\%$

Slope for 10kΩ $= \frac{Y_2 - Y_1}{X_2 - X_1} = \frac{0.7 - 0.3}{7 - 3} = 0.1$

$\therefore$ Resistance $= \frac{1}{\text{Slope for } 10\,k\Omega}$

$= \frac{1}{0.1}$

$= 10$

Measured R was $= 9.85\ k\Omega$

$\therefore$ Error $= \frac{10 - 9.85}{10} \times 100\%$

$= 1.5\%$